Package 'stratsampling'

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Title Stratified Sampling for Model-Assisted Treatment Effect Estimation

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Description Tools for stratified sampling designs that leverage surrogate scores (e.g., machine learning predictions) to efficiently estimate treatment effects when human coding is expensive. Provides functions for creating and evaluating candidate stratifications, drawing stratified samples, and computing model-assisted estimators that combine surrogate predictions with a bias-corrected adjustment from a coded subsample. Also includes power analysis tools and simulation utilities for comparing sampling designs. Methods are described in Mozer, Pashley, and Miratrix (2026).

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URL https://lmiratrix.github.io/stratsampling/, https://github.com/lmiratrix/stratsampling

BugReports https://github.com/lmiratrix/stratsampling/issues

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`calc_nominal_var` *Estimate what the SE would be given a stratification design*

Description

This method calculates the SE given some possible population summary statistics for the proportional stratified estimator. This method is used for making power calculations when planning to subsample documents.

Usage

```
calc_nominal_var(N, p, h, var_Y0, R2_ML, R2_strat)
```

Arguments

`N`	The total population size
`p`	The proportion of units assigned to treatment
`h`	The proportion of units that will be human coded (i.e., the sampling fraction)
`var_Y0`	The variance of the outcome in the control group

R2_ML	The pseudo R-squared of the model's surrogate outcomes for predicting actual outcomes.
R2_strat	The proportion of the residual variance explained by the stratification variable.

Value

The estimated variance of the model-assisted proportional stratified estimator.

`calc_optimal_allocation_oracle` *Oracle Optimal (Neyman) Allocation Calculator*

Description

Computes the optimal sample allocation across strata using Neyman allocation, which allocates more units to strata with higher residual variance. This is an oracle method because it requires knowing the true outcomes (Yobs) for all units in order to compute the actual residuals.

Usage

```
calc_optimal_allocation_oracle(
  dat,
  n_code,
  Z = NULL,
  strata = NULL,
  Yobs = "Yobs",
  Yhat = "Yhat",
  return_counts = FALSE,
  seed = NULL
)
```

Arguments

dat	Data frame containing the full population with observed outcomes and surrogate predictions.
n_code	Integer. Total number of units to sample for coding.
Z	Character string naming the treatment indicator column, or NULL to ignore treatment. Default is NULL.
strata	Character string naming the stratification column, or NULL (single stratum). Default is NULL.
Yobs	Character string or vector. The observed outcome variable. Default is "Yobs".
Yhat	Character string or vector. The surrogate prediction variable. Default is "Yhat".
return_counts	Logical. If TRUE, returns only the stratum-level allocation table instead of the full dataset with sampling indicators. Default is FALSE.
seed	Optional integer seed for reproducibility.

Value

If return_counts = FALSE (default), returns the input data frame augmented with sampling indicator S and sampling probability pi. If return_counts = TRUE, returns a data frame with the optimal allocation counts per stratum.

See Also

generate_sample_index for proportional allocation, est_ma_strat for the stratified model-assisted estimator.

`calc_sampling_weights` *Calculate Sampling Weights from a Stratified Sample*

Description

Given a data frame with a sampling indicator and (optional) stratification, computes the inclusion probability for each unit as the proportion sampled within its stratum (and treatment arm, if specified).

Usage

```
calc_sampling_weights(dat, Z = NULL, S = "S", strata = NULL)
```

Arguments

`dat`	Data frame containing the population.
`Z`	Character string naming the treatment indicator column, or `NULL` to ignore treatment. Default is `NULL`.
`S`	Character string naming the sampling indicator column (values 0/1), or a numeric vector of indicators. Default is `"S"`.
`strata`	Character string naming the stratification column, or `NULL` for unstratified weights. Default is `NULL`.

Value

Numeric vector of sampling probabilities (one per row of `dat`), where each value equals n_k / N_k for the unit's stratum.

See Also

generate_sample_index, est_ma_strat

`compare_decomposition_oracle` *Compare Candidate Stratifications Using Oracle Variance Decomposition*

Description

For each candidate stratification, computes the BS (between-strata) and WS (within-strata) variance components from Theorem 3.2, which together determine the variance reduction from stratified sampling. Returns a summary table and visualization comparing candidates based on oracle information (true residuals).

Usage

```
compare_decomposition_oracle(data, candidates, n_code, Z = "Z")
```

Arguments

`data`	Data frame containing `Yobs`, `Yhat`, and optionally a treatment indicator column.
`candidates`	Named list of candidate stratification vectors. Each element should be a factor or character vector of the same length as `nrow(data)` containing stratum assignments.
`n_code`	Total number of units to sample for human coding. Required for computing the exact Theorem 3.2 formulas.
`Z`	Character string specifying the name of the treatment indicator column, or `NULL` to ignore treatment groups. Default is `"Z"`.

Details

This function uses oracle information (the true residuals Yobs - Yhat) to compute the exact variance decomposition for each candidate stratification. The key metric is BS - WS, which represents the variance reduction achieved by stratified sampling compared to simple random sampling (Theorem 3.2):

$$Var(\hat{\tau}_{SRS}) - Var(\hat{\tau}_{strat}) = BS - WS$$

Stratifications with larger BS - WS values provide greater variance reduction. The BS component measures systematic differences in prediction bias across strata, while the WS component (typically small under proportional allocation) reflects within-stratum residual heterogeneity.

Value

A list with three elements:

`summary` Data frame with one row per stratification containing:

- `stratification`: Name of the stratification
- `n_strata`: Number of strata
- `BS`: Between-strata variance component
- `WS`: Within-strata variance component
- `BS_minus_WS`: Variance reduction (BS - WS)
- `var_ratio`: Ratio of max to min within-stratum variance
- `rank`: Rank by `BS_minus_WS` (1 = best)

`decomposition_details` List of full decomposition results for each stratification (output from `compute_variance_decomposition`).

See Also

`compute_variance_decomposition` for the underlying decomposition.

Examples

```
## Not run:
# Create candidate stratifications
candidates <- create_stratification_candidates(data, vars = c("Yhat", "word_count"))

# Compare using oracle decomposition
results <- compare_decomposition_oracle(data, candidates, n_code = 200)

# View summary table
results$summary

## End(Not run)
```

`compare_oracle_vs_precoding` *Compare Pre-coding Proxy Metrics with Oracle Metrics*

Description

Creates a combined table and visualization comparing the pre-coding proxy metrics (based only on Yhat) with oracle metrics (based on actual residuals) for multiple candidate stratifications. This validates whether the pre-coding metrics successfully identify stratifications that perform well in practice.

Usage

```
compare_oracle_vs_precoding(
  oracle_comparison,
  precoding_comparison,
  top_n = 10
)
```

Arguments

`oracle_comparison`	Output from `compare_stratifications`, which uses actual residuals (Yobs - Yhat).
`precoding_comparison`	Output from `compare_stratifications_precoding`, which uses only Yhat as a proxy.
`top_n`	Integer specifying how many top stratifications to include in the output. Default is 10.

Details

This function is useful for validating that the pre-coding proxy metrics (which can be computed before any human coding is done) successfully identify stratifications that would perform well according to the oracle metrics (which require knowing the true residuals).

The key comparison is between:

- `mean_spread` (oracle): Variance of mean residuals across strata (BS component)

- mean_spread (pre-coding): Variance of mean Yhat across strata (proxy for BS)

High correlation between rankings suggests the pre-coding metrics are reliable for selecting stratifications in practice.

Value

A list with three elements:

combined_table Data frame merging oracle and pre-coding metrics for each stratification, with columns for both metric types and rank comparisons.

correlation Spearman correlation between oracle and pre-coding rankings, measuring how well pre-coding metrics predict oracle performance.

plot A ggplot object showing the relationship between pre-coding and oracle metrics.

See Also

compare_stratifications, compare_stratifications_precoding

---

compare_stratifications

Compare Multiple Candidate Stratification Variables

Description

Evaluates and compares multiple candidate stratification variables to help select the best one for stratified model-assisted estimation. Each candidate is assessed using evaluate_stratification, and results are summarized in a ranked table.

Usage

```
compare_stratifications(data, candidates, Z = "Z", ...)
```

Arguments

data	Data frame containing a resid column (residuals = Yobs - Yhat), a treatment indicator column, and any columns referenced by the candidate stratifications.
candidates	Named list of candidate stratification vectors. Each element should be a vector of the same length as nrow(data) containing stratum assignments. Names are used to identify stratifications in the output.
Z	Character string. Name of the treatment indicator column in data. Default is "Z".
...	Additional arguments passed to evaluate_stratification.

Value

A data frame of performance metrics for each stratification method, sorted by promise.

compare_stratifications_precoding
Compare candidate stratifications using pre-coding metrics

Description

Compare candidate stratifications using pre-coding metrics

Usage

```
compare_stratifications_precoding(data, candidates, Z = NULL, Yhat = "Yhat")
```

Arguments

data	Data frame with Yhat and Z columns
candidates	Named list of stratification vectors
Z	Character string specifying the name of the treatment indicator or NULL to ignore treatment groups.
Yhat	Character string specifying the name of the column containing predicted values (Yhat). Default is "Yhat".

Value

Data frame comparing stratifications

compute_variance_decomposition
Compute variance decomposition (BS and WS components)

Description

Computes the between-strata (BS) and within-strata (WS) variance components of the residuals. These components are related to the variance reduction from stratified sampling as described in Theorem 3.2 of the manuscript.

Usage

```
compute_variance_decomposition(dat, Z = "Z", strata = NULL, n_code = NULL)
```

Arguments

dat	Data frame with Yobs, Yhat columns
Z	Character string specifying the treatment indicator column name, or NULL to ignore treatment groups. Default is "Z".
strata	Stratification variable. Can be a vector of stratum assignments (same length as nrow(dat)) or NULL if dat already contains a column named "strata".
n_code	Optional. Total number of units to be sampled for coding. If provided, the function computes the exact BS and WS terms from Theorem 3.2 which depend on the sampling fraction. If NULL (default), computes simplified population-level variance decomposition components that serve as proxies.

Details

The variance reduction from stratified sampling (Theorem 3.2) is:

$$Var(\hat{\tau}_{SRS}) - Var(\hat{\tau}_{strat}) = BS - WS$$

Simplified (population-level) decomposition (when `n_code = NULL`):

This computes the standard ANOVA-style decomposition of residual variance:

- `BS = sum_k w_k * (mean_resid_k - grand_mean)^2`
- `WS = sum_k w_k * var_resid_k`

where `w_k = N_zk / N_z` is the stratum weight. This satisfies:

$$Var(resid) = BS + WS$$

These simplified components serve as useful diagnostics:

- Large BS suggests systematic differences in prediction bias across strata, which stratification can exploit.
- Heterogeneous within-stratum variances (captured by `var_ratio`) suggest potential benefit from optimal allocation.

Exact Theorem 3.2 decomposition (when `n_code` is provided):

The exact formulas incorporate finite population corrections and sampling fractions. The BS term from Theorem 3.2 is:

$$BS = sum_k[(N_z - n_z)/N_z * N_zk/(n_z(N_z - 1)) * (mean_resid_k - grand_mean)^2]$$

The WS term is more complex and depends on the sampling design. Under proportional allocation, WS is approximately zero. Under optimal allocation (oversampling high-variance strata), WS can become negative, providing additional variance reduction.

The returned decomposition includes:

`BS` Between-strata variance component. Measures how much stratum mean residuals differ from the grand mean. Larger values indicate greater potential benefit from stratification.

`WS` Within-strata variance component. Weighted average of within-stratum residual variances. Under proportional allocation, this term is typically small and positive. NOTE: The exact WS term from Theorem 3.2 can be negative under optimal allocation (oversampling high-variance strata), but the simplified population version computed here is always positive.

`BS_minus_WS` The difference BS - WS, which approximates the variance reduction from stratified sampling vs. SRS. Positive values indicate stratification reduces variance.

Value

If `Z` is specified (treatment/control), returns a list with:

`by_arm` Data frame with decomposition for each treatment arm.

`overall` Data frame with overall decomposition (summed over arms).

If `Z` is `NULL`, returns a single data frame with the decomposition. See details for the specific components included in each case.

See Also

`evaluate_stratification` for computing stratification quality metrics, `plot_strata_residuals` for visualization.

`create_stratification_candidates` *Create Candidate Stratification Variables*

Description

Generates a set of candidate stratification variables for use in stratified model-assisted estimation. Creates single-variable stratifications (tertiles, quartiles, quintiles) for each specified variable, as well as interaction stratifications for all pairs of variables.

Usage

```
create_stratification_candidates(
  dat,
  vars,
  Z = NULL,
  jitter_vars = NULL,
  sd_jitter = 0.001,
  seed = NULL
)
```

Arguments

`dat`	Data frame containing the variables to stratify on.
`vars`	Character vector specifying the names of columns in `dat` to use for creating stratifications.
`Z`	Optional character string specifying the name of the treatment indicator variable in `dat`. If provided, stratifications are computed separately within each treatment arm, ensuring balanced strata across arms. Default is `NULL` (pooled stratification).
`jitter_vars`	Character vector specifying which variables (subset of `vars`) should have small random noise added before computing quantile breaks. This helps avoid ties that can cause `cut()` to fail. Default is `NULL` (no jitter applied).
`sd_jitter`	Numeric value specifying the standard deviation of the jitter noise. Default is `1e-3`.
`seed`	Optional integer seed for reproducibility when jitter is applied. Default is `NULL`.

Details

For each variable in `vars`, the function creates three stratifications based on quantiles:

- Tertiles (3 groups): Low, Med, High
- Quartiles (4 groups): Low, Med-Low, Med-High, High
- Quintiles (5 groups): Very Low, Low, Med, High, Very High

For each unique pair of variables, the function creates four interaction stratifications by crossing quantile groups:

- 2x2: median split on both variables (4 strata)
- 2x3: median split on first, tertiles on second (6 strata)

- 3x2: tertiles on first, median split on second (6 strata)
- 3x3: tertiles on both variables (9 strata)

When `Z` is provided, quantile breaks are computed separately within each treatment arm. This ensures that the same proportion of treated and control units fall into each stratum level, which is important for maintaining balance in stratified estimation.

When a variable has many tied values, quantile breaks may not be unique, causing `cut()` to fail. Use `jitter_vars` to add small random noise (SD = 0.001) to such variables before computing breaks.

Value

A named list of factor vectors, each representing a candidate stratification. Names follow the convention:

Single-variable `{var}_tertiles`, `{var}_quartiles`, `{var}_quintiles`

Interactions `{var1}_{var2}_2x2`, `{var1}_{var2}_2x3`, `{var1}_{var2}_3x2`, `{var1}_{var2}_3x3`

Factor levels have meaningful labels (e.g., "Low Yhat", "Med Yhat / High word_count"). Stratifications that fail due to non-unique quantile breaks are automatically excluded from the output.

`describe_population` *Describe the Data-Generating Process Population*

Description

Utility function to explore how different simulation factors (bias pattern, residual variance heterogeneity, R-squared) interact to produce the population-level properties of the surrogate predictions. Generates a large population (or accepts an existing data frame) and summarizes the variance of bias, average residual variance, proportion of error due to bias, and achieved R-squared.

Usage

```
describe_population(
  n_strat,
  strata_config = c("balanced-exact", "balanced-approx", "unbalanced"),
  tau = 0,
  bias = c("none", "small", "moderate", "large", "extreme-contrast"),
  residual_var = c("homogeneous", "heterogeneous", "extreme-contrast"),
  sd_y = 3,
  R2_target = 0.5,
  no_tx = FALSE,
  rescale_v = TRUE,
  seed = NULL,
  verbose = FALSE
)
```

Arguments

n_strat	Integer specifying the number of strata, or a data frame (output of generate_population) to summarize directly.
strata_config	Character. How to allocate units across strata: "balanced-exact" (equal sizes, requires N divisible by n_strat), "balanced-approx" (approximately equal via random assignment), or "unbalanced" (random proportions).
tau	Numeric. Constant additive treatment effect. Default is 0 (no treatment effect).
bias	Character. Level of systematic prediction bias across strata: "none", "small" (up to 25% of sd_y), "moderate" (up to 50%), "large" (up to 100%), or "extreme-contrast" (bias concentrated in two strata).
residual_var	Character. Pattern of residual variance across strata: "homogeneous" (equal variance), "heterogeneous" (linearly increasing), or "extreme-contrast" (one very high, one very low).
sd_y	Numeric. Standard deviation of the outcome variable. Default is 3.
R2_target	Numeric between 0 and 1. Target R-squared for the surrogate predictions. Controls overall prediction quality. Default is 0.5.
no_tx	Logical. If TRUE, do not generate treatment assignment; all units receive control potential outcomes. Useful for single-arm designs. Default is FALSE.
rescale_v	Logical. If TRUE, rescale residual variances to preserve the target R-squared exactly. Default is TRUE.
seed	Optional integer seed for reproducibility.
verbose	Logical. If TRUE, prints formatted tables of stratum-level and aggregate summaries. Default is FALSE.

Value

A one-row tibble with columns: Z (treatment arm), Var_b (variance of stratum-level biases), E_sig2 (expected within-stratum residual variance), prop_bias (proportion of prediction error due to bias), and R2 (achieved R-squared of surrogate predictions).

See Also

generate_population

---

estimate_R2 *Given data, estimate the R2 of the stratification and ML*

Description

This can be used on pilot data to get a rough sense of values for a power calculation.

Usage

```
estimate_R2(
  data = NULL,
  Yhat = "Yhat",
  Yobs = "Yobs",
  S = NULL,
  Z = "Z",
  strata = "strata"
)
```

Arguments

`data`	Data frame containing outcome, surrogate, sampling indicator, treatment, and stratification columns.
`Yhat`	Name of the surrogate score column. Default is `"Yhat"`.
`Yobs`	Name of the human-coded outcome column. Default is `"Yobs"`.
`S`	Name of the sampling indicator column (1 = coded, 0 = not coded). Default is `"S"`.
`Z`	Name of the treatment indicator column, or `NULL` for a population mean. Default is `"Z"`.
`strata`	Name of the stratification column, or `NULL` for unstratified (SRS) estimation. Default is `"strata"`.

Details

This is a back of the envelope calculation that does not take varying proportions of human coded outcomes within strata into account.

Value

A tibble containing the estimated R2 of the stratification (how much machine predictions vary across strata) and of the surrogates for the outcome.

See Also

`calc_nominal_var()`, `plot_power_curve()`, `make_power_table()`

`est_ma_srs` *Model-Assisted SRS Estimator*

Description

Computes the model-assisted treatment effect estimator using a simple random subsample (no stratification). This is a convenience wrapper around `est_ma_strat()` with `strata = NULL`.

Usage

```
est_ma_srs(
  data,
  Yhat = "Yhat",
  Yobs = "Yobs",
  S = "S",
  Z = "Z",
  finite_pop = FALSE
)
```

Arguments

data	Data frame containing outcome, surrogate, sampling indicator, treatment, and stratification columns.
Yhat	Name of the surrogate score column. Default is "Yhat".
Yobs	Name of the human-coded outcome column. Default is "Yobs".
S	Name of the sampling indicator column (1 = coded, 0 = not coded). Default is "S".
Z	Name of the treatment indicator column, or NULL for a population mean. Default is "Z".
finite_pop	Logical. If TRUE, applies finite population corrections and returns only the sampling variance component. Default is FALSE.

Value

A data frame with the same structure as est_ma_strat().

See Also

est_ma_strat() for the stratified version, est_subset() for the baseline estimator.

---

est_ma_strat *Model-Assisted Stratified Estimator*

Description

Computes the model-assisted treatment effect estimator that combines surrogate predictions for the full population with a bias correction from a human-coded stratified subsample. Within each treatment arm and stratum, the estimator adjusts the population mean of surrogate scores by the average residual (human score minus surrogate) in the coded units, weighted by stratum sizes.

Usage

```
est_ma_strat(
  data = NULL,
  Yhat = "Yhat",
  Yobs = "Yobs",
  S = "S",
  Z = "Z",
  strata = "strata",
  return_pieces = FALSE,
  finite_pop = FALSE
)
```

Arguments

data	Data frame containing outcome, surrogate, sampling indicator, treatment, and stratification columns.
Yhat	Name of the surrogate score column. Default is "Yhat".
Yobs	Name of the human-coded outcome column. Default is "Yobs".

S	Name of the sampling indicator column (1 = coded, 0 = not coded). Default is "S".
Z	Name of the treatment indicator column, or NULL for a population mean. Default is "Z".
strata	Name of the stratification column, or NULL for unstratified (SRS) estimation. Default is "strata".
return_pieces	Logical. If TRUE, returns a list with intermediate quantities useful for diagnostics. Default is FALSE.
finite_pop	Logical. If TRUE, applies finite population corrections and returns only the sampling variance component. Default is FALSE.

Details

If strata is set to NULL, this reduces to the SRS model-assisted estimator (equivalent to calling est_ma_srs()).

Value

A data frame with columns:

est Point estimate of the treatment effect (or mean).

SE Oracle SE computed using true residual variances for all units.

SEhat Estimated SE from the coded subsample, combining both the full-coding and sampling variance components.

SEhat.full Estimated SE component attributable to treatment effect variability (i.e., variance even if all units were coded).

SEhat.samp Estimated SE component attributable to subsampling (i.e., additional variance from coding only a subset).

S2.0, S2.1 Estimated population variance of outcomes in control and treatment groups.

See Also

est_ma_srs() for the unstratified version, est_subset() for the baseline estimator, and generate_sample_index() for drawing stratified samples.

---

est_oracle *Oracle Estimate (Full Coding Benchmark)*

Description

Computes the treatment effect estimate assuming all units have been human-coded. This serves as a benchmark for evaluating the efficiency of subsampling designs.

Usage

```
est_oracle(data, Yobs = "Yobs", Z = "Z")
```

Arguments

data	Data frame containing at minimum the outcome column.
Yobs	Name of the observed outcome column, or a numeric vector. Default is "Yobs".
Z	Name of the treatment indicator column, or NULL for estimating a population mean. Default is "Z".

Value

A data frame with columns method, est (point estimate), and SE (standard error).

Examples

```
data(toy_data_coded)
# Oracle is usually computed on fully-coded data
# est_oracle(full_data, Yobs = "Yobs", Z = "Tx")
```

---

est_subset *Simple Subset Estimator (Baseline)*

Description

Computes the naive difference-in-means treatment effect estimator using only the human-coded subsample, ignoring surrogate scores entirely. This is the baseline estimator that the model-assisted estimators aim to improve upon.

Usage

```
est_subset(data, Yobs = "Yobs", S = "S", Z = "Z", finite_pop = FALSE)
```

Arguments

data	Data frame containing at minimum the outcome, sampling indicator, and treatment columns.
Yobs	Name of the observed outcome column, or a numeric vector. Default is "Yobs".
S	Name of the sampling indicator column (1 = coded, 0 = not coded), or a numeric vector. Default is "S".
Z	Name of the treatment indicator column, or NULL for estimating a population mean. Default is "Z".
finite_pop	Logical. If TRUE, applies a finite population correction. Default is FALSE.

Value

A data frame with columns:

est Point estimate of the treatment effect (or mean).

SE Oracle SE (computed using full population variance).

SEhat Estimated SE from the coded subsample.

SEhat.full NA (not applicable for this estimator).

SEhat.samp NA (not applicable for this estimator).

S2.0, S2.1 Sample variance of coded outcomes in control and treatment groups, respectively.

See Also

`est_ma_srs()` and `est_ma_strat()` for model-assisted alternatives that leverage surrogate scores.

`evaluate_stratification` *Evaluate a Candidate Stratification Variable*

Description

Computes stratum-level residual statistics and heterogeneity metrics to assess the potential benefit of a stratification for model-assisted estimation.

Usage

```
evaluate_stratification(data, strata_var, Z = "Z", min_var_threshold = 0.01)
```

Arguments

`data`	Data frame containing a `resid` column (residuals = Yobs - Yhat) and the stratification variable specified by `strata`.
`strata_var`	Character string specifying the name of the column to use as the stratification variable.
`Z`	Character string specifying the name of the treatment indicator or `NULL` to ignore treatment groups.
`min_var_threshold`	Numeric. Minimum within-stratum residual variance required. If any stratum has variance below this threshold, the function returns `NULL`. This guards against strata with near-zero variance that could cause numerical issues. Default is `0.01`.

Details

This function is useful for comparing candidate stratification variables before selecting one for stratified sampling. Key metrics to examine:

- `var_mean_resid`: Higher values indicate greater between-strata variance (BS), suggesting stratification will reduce sampling variance.
- `ratio_var`: Values greater than 2 indicate heteroskedasticity across strata, suggesting optimal (Neyman) allocation will outperform proportional allocation.
- `imbalance`: Values close to 1 indicate balanced strata; highly unbalanced strata may reduce efficiency.

Value

A list with three elements, or `NULL` if the stratification is invalid (e.g., empty data, missing values, or variance below threshold):

`strata_stats` Data frame with stratum-level statistics:

- `N_zk` (or `N_k`): Sample size in each stratum
- `mean_resid`: Mean residual in each stratum

- `var_resid`: Variance of residuals in each stratum
- `sd_resid`: Standard deviation of residuals in each stratum

If `Z` is specified, includes treatment indicator `Z`.

`heterogeneity` Data frame summarizing heterogeneity across strata:

- `n_strata`: Number of strata
- `min_size`, `max_size`: Range of stratum sizes
- `imbalance`: Ratio of largest to smallest stratum size
- `min_var`, `max_var`: Range of residual variances
- `ratio_var`: Ratio of max to min residual variance (heteroskedasticity measure)
- `range_var`: Difference between max and min residual variance
- `var_mean_resid`: Variance of stratum mean residuals (proxy for BS)

If `Z` is specified, returns one row per treatment arm.

`stratification` Character string echoing the stratification variable name.

`evaluate_stratification_precoding` *Evaluate a candidate stratification strategy using pre-coding proxies*

Description

Since we don't have residuals, we use proxies:

- Variance of Yhat within strata (lower = more homogeneous predictions)
- Spread of stratum means (higher = better potential for bias correction)
- Balance of stratum sizes

Usage

```
evaluate_stratification_precoding(data, strata = NULL, Z = "Z", Yhat = "Yhat")
```

Arguments

`data`	Data frame with Yhat column
`strata`	Stratification vector
`Z`	Character string specifying the name of the treatment indicator or `NULL` to ignore treatment groups.
`Yhat`	Character string specifying the name of the column containing predicted values (Yhat). Default is `"Yhat"`.

Value

Data frame with proxy quality metrics

generate_observed_sample *Generate Observed Dataset for Simulation*

Description

Wrapper around generate_sample_index that draws a (stratified) sample and sets unsampled outcomes to NA. This simulates the real-world scenario where only a subsample of units receives human coding.

Usage

```
generate_observed_sample(
  dat,
  n_code,
  Z = NULL,
  strata = NULL,
  by_tx = FALSE,
  seed = NULL
)
```

Arguments

dat	Data frame containing the population.
n_code	Sample size specification. If a single number, units are allocated proportionally across strata (and treatment arms if Z is specified). If a vector matching the number of strata-by-treatment cells, specifies exact counts per cell.
Z	Character string naming the treatment indicator column, or NULL to ignore treatment assignment. When specified, sampling is done within substrata defined by crossing strata with treatment. Default is NULL.
strata	Character string naming the stratification column, or NULL for SRS (all units in one stratum). Default is NULL.
by_tx	Logical. If TRUE, sample separately within treatment arms. Default is FALSE.
seed	Optional integer seed for reproducibility.

Details

This function is intended for simulation studies only; it requires fully observed outcomes for the entire population.

Value

The input data frame with two additional/modified columns:

S Sampling indicator (1 = coded, 0 = not coded)

Yobs Observed outcome; set to NA for unsampled units

See Also

generate_sample_index, generate_population

`generate_population` *Generate a Simulated Population with Surrogate Scores*

Description

Creates a simulated population with potential outcomes, treatment assignment, and surrogate predictions (Yhat) that exhibit configurable levels of bias and residual variance heterogeneity across strata. This is the primary data-generating function for simulation studies comparing stratified vs. SRS sampling designs.

Usage

```
generate_population(
  N,
  n_strat,
  strata_config = c("balanced-exact", "balanced-approx", "unbalanced"),
  tau = 0,
  bias = c("none", "small", "moderate", "large", "extreme-contrast"),
  residual_var = c("homogeneous", "heterogeneous", "extreme-contrast"),
  sd_y = 3,
  R2_target = 0.5,
  no_tx = FALSE,
  rescale_v = TRUE,
  seed = NULL
)
```

Arguments

`N`	Integer. Total population size.
`n_strat`	Integer. Number of strata to create within each treatment arm.
`strata_config`	Character. How to allocate units across strata: `"balanced-exact"` (equal sizes, requires N divisible by n_strat), `"balanced-approx"` (approximately equal via random assignment), or `"unbalanced"` (random proportions).
`tau`	Numeric. Constant additive treatment effect. Default is 0 (no treatment effect).
`bias`	Character. Level of systematic prediction bias across strata: `"none"`, `"small"` (up to 25% of sd_y), `"moderate"` (up to 50%), `"large"` (up to 100%), or `"extreme-contrast"` (bias concentrated in two strata).
`residual_var`	Character. Pattern of residual variance across strata: `"homogeneous"` (equal variance), `"heterogeneous"` (linearly increasing), or `"extreme-contrast"` (one very high, one very low).
`sd_y`	Numeric. Standard deviation of the outcome variable. Default is 3.
`R2_target`	Numeric between 0 and 1. Target R-squared for the surrogate predictions. Controls overall prediction quality. Default is 0.5.
`no_tx`	Logical. If `TRUE`, do not generate treatment assignment; all units receive control potential outcomes. Useful for single-arm designs. Default is `FALSE`.
`rescale_v`	Logical. If `TRUE`, rescale residual variances to preserve the target R-squared exactly. Default is `TRUE`.
`seed`	Optional integer seed for reproducibility.

Details

Strata are generated post-treatment: `n_strat` strata are created within treated and control units separately, and surrogate noise (bias + residual) is added within each stratum.

Value

A data frame with columns:

Z Treatment indicator (0 = control, 1 = treated)

strata Stratum assignment (integer 1 to n_strat)

ID Unique unit identifier

Y0 Control potential outcome

Y1 Treated potential outcome (Y0 + tau)

Yobs Observed outcome under assigned treatment

Yhat Surrogate prediction (Yobs + stratum-specific noise)

See Also

`describe_population` for summarizing the DGP, `generate_observed_sample` for drawing a subsample.

Examples

```
dat <- generate_population(N = 400, n_strat = 4, tau = 0.5,
                           bias = "moderate",
                           residual_var = "heterogeneous",
                           R2_target = 0.7, seed = 42)
head(dat)
table(dat$strata, dat$Z)
```

---

generate_sample_index *Generate Stratified or Simple Random Sample Indicators*

Description

Given a population data frame, generates a binary vector indicating which units are selected for human coding. Supports simple random sampling (SRS), stratified proportional allocation, and custom per-stratum sample sizes.

Usage

```
generate_sample_index(
  dat,
  n_code,
  Z = NULL,
  strata = NULL,
  p_tx = 0.5,
  seed = NULL
)
```

Arguments

dat	Data frame containing the population.
n_code	Sample size specification. If a single number, units are allocated proportionally across strata (and treatment arms if Z is specified). If a vector matching the number of strata-by-treatment cells, specifies exact counts per cell.
Z	Character string naming the treatment indicator column, or NULL to ignore treatment assignment. When specified, sampling is done within substrata defined by crossing strata with treatment. Default is NULL.
strata	Character string naming the stratification column, or NULL for SRS (all units in one stratum). Default is NULL.
p_tx	Numeric between 0 and 1. Proportion of the total sample allocated to the treatment arm when Z is specified and n_code is a single number. If NULL, uses the observed treatment proportion. Default is 0.5.
seed	Optional integer seed for reproducibility.

Value

An integer vector of length nrow(dat) with values 0 (not sampled) or 1 (sampled).

See Also

generate_observed_sample for a wrapper that also masks unsampled outcomes, calc_sampling_weights for computing the implied sampling probabilities.

---

make_interaction_strata

Create Interaction-Based Stratification (Continuous or Categorical)

Description

Constructs an interaction stratification from two variables. Each variable may be continuous (binned into quantile-based strata) or categorical (used directly as factor levels). Optionally, quantile strata can be constructed within treatment arms.

Usage

```
make_interaction_strata(
  x1,
  n1,
  var_name1 = NULL,
  x2 = NULL,
  n2 = NULL,
  var_name2 = NULL,
  z_vec = NULL
)
```

Arguments

x1	First stratification variable.
n1	Number of quantile strata for x1 (ignored if categorical).
var_name1	Character string used in labeling strata for x1.
x2	Second stratification variable.
n2	Number of quantile strata for x2 (ignored if categorical).
var_name2	Character string used in labeling strata for x2.
z_vec	Optional treatment indicator. If supplied and a variable is continuous, quantile strata are constructed within treatment arms.

Details

If a variable is a factor, character, or logical vector—or if n is NULL or NA—it is treated as categorical. Otherwise, it is treated as continuous and partitioned into n quantile strata.

Value

A factor representing the interaction of the two stratifications, with empty levels dropped. Returns NULL if either stratification fails.

---

make_power_table *Power curve table and plot for stratified vs SRS sampling*

Description

Build the power-curve table (via make_power_table()) and/or a ggplot (via plot_power_curve()) comparing stratified sampling to SRS, given population parameters.

Usage

```
make_power_table(N, p, var_Y0, R2_ML, R2_strat, h_min = 0.05)

plot_power_curve(N, p, var_Y0, R2_ML, R2_strat, h_min = 0.05)
```

Arguments

N	The total population size
p	The proportion of units assigned to treatment
var_Y0	The variance of the outcome in the control group
R2_ML	The pseudo R-squared of the model's surrogate outcomes for predicting actual outcomes.
R2_strat	The proportion of the residual variance explained by the stratification variable.
h_min	Minimum proportion of documents coded to include in the plot/table.

Details

You can estimate parameters from pilot data with estimate_R2().

Value

`make_power_table()` returns a tibble with columns h, SE, and MDES.

`plot_power_curve()` returns a ggplot object with power curves. The underlying data table is stored as an attribute `"table"`.

See Also

`calc_nominal_var()` for calculating for a single scenario and fixed h, and `estimate_R2()` for estimating design parameterd.

`plot_decomposition_comparison` *Plot Oracle Decomposition Comparison*

Description

Creates a horizontal bar chart comparing the variance reduction (BS - WS) across candidate stratifications, with bars colored by the relative contribution of BS vs WS.

Usage

```
plot_decomposition_comparison(
  decomposition_result,
  top_n = NULL,
  show_components = FALSE
)
```

Arguments

`decomposition_result`	Output from `compare_decomposition_oracle`.
`top_n`	Integer specifying how many top stratifications to display. Default is `NULL` (show all).
`show_components`	Logical. If `TRUE`, shows separate bars for BS and WS. If `FALSE` (default), shows only BS - WS as a single bar.

Value

A ggplot object.

plot_strata_residuals *Create visualization of residual structure by strata*

Description

Create visualization of residual structure by strata

Usage

```
plot_strata_residuals(dat, strata)
```

Arguments

dat	Data frame with Yobs, Yhat, Z, strata columns
strata	Stratification variable

Value

ggplot object

toy_data *Toy dataset for stratified sampling examples*

Description

A simulated dataset containing surrogate predictions and baseline covariates for 1000 observations from a randomized controlled trial. This dataset is used in the package vignettes to demonstrate the stratification, sampling, and estimation workflow.

Usage

```
toy_data
```

Format

A data frame with 1000 rows and 6 columns:

X1 Baseline covariate 1 (continuous).

X2 Baseline covariate 2 (continuous).

X3 Baseline covariate 3 (continuous). In this toy example, X3 moderates the accuracy of the surrogate predictions.

Yhat Surrogate prediction of the outcome, e.g., from a machine learning model applied to the full sample.

Tx Treatment assignment indicator (0 = control, 1 = treatment).

ID Unique observation identifier.

Examples

```
data(toy_data)
head(toy_data)
```

`toy_data_coded` *Coded outcomes for the toy dataset subsample*

Description

A subset of `toy_data` containing human-coded outcomes for the observations selected for coding. This dataset is used in the package vignettes to demonstrate merging coded outcomes back into the full dataset for model-assisted estimation.

Usage

```
toy_data_coded
```

Format

A data frame with approximately 100 rows and 2 columns:

ID Unique observation identifier, matching `toy_data$ID`.

Yobs Human-coded outcome for the selected observation.

Examples

```
data(toy_data_coded)
head(toy_data_coded)
```

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